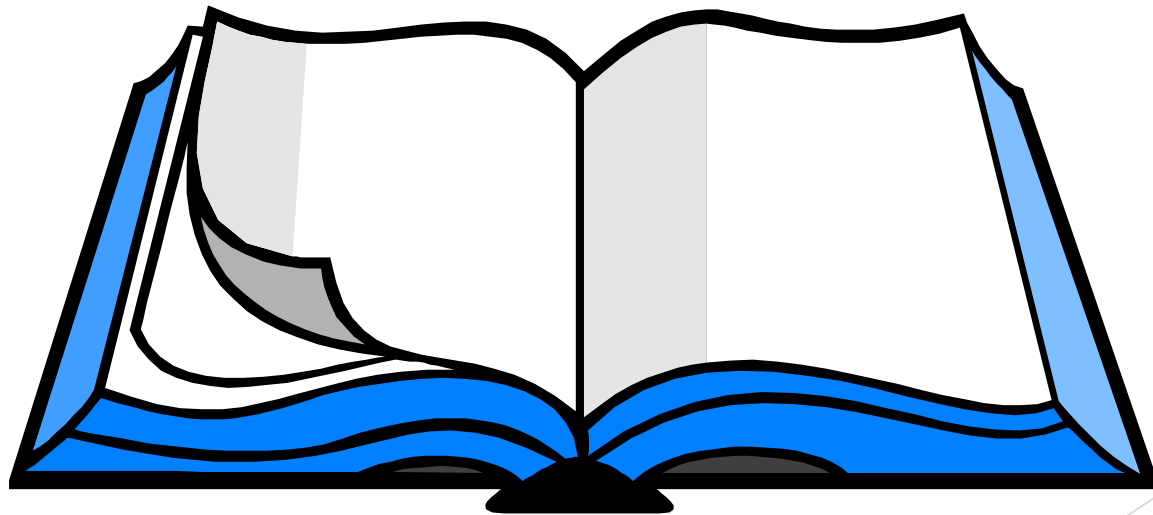


Numerical Methods Error Analysis



Types of Error

- Errors are of two types:

1. Truncation Error
2. Round-off Error

- **Truncation error** is a result of using approximations to represent exact mathematical procedures.
- **Round-off error** occurs when only certain digits and decimal places are used to represent exact numbers.

Example 1:

1. We know that

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - + \dots$$

Suppose we use it to evaluate $\sin 30^\circ$.

$$x = 30^\circ = \pi/6 \approx 0.5236$$

Our **first approximation** is

$$\sin 30^\circ \approx 0.5236.$$

Second approximation:

$$\sin 30^\circ \approx 0.5236 - 0.5236^3/6 \\ = 0.499675$$

Third approximation:

$$\sin 30^\circ \approx 0.499675 + 0.5236^5/120 \\ = 0.5000029570$$

The error involved in the approximations here is **truncation error**, since terms in the approximating series are truncated.

Example 2:

- Consider the number 3.12746
- If the number is written correct to 3 decimal places, then it is approximated as 3.127
- If the number is written correct to 4 decimal places, then it is approximated as 3.1275
- The error involved in this kind of approximation is called **round-off error**.

Measurement of Errors:

- Precision and accuracy are important criteria for the assessment of approximate solutions.
- There are formulae to measure these characteristics.
- When the current solution is compared with the true solution, the error involved is called **true error**.
- When the current solution is compared with the solution obtained in the previous iteration, the error involved is called **approximate error**.

1. True Error:

- It is used to measure the lack of accuracy of an estimate.
- True (absolute) error = E_t
= True value – Approximation
- True relative error = E_t / True value

True percent relative error = ε_t

$$= \frac{\text{True value} - \text{approximation}}{\text{True value}} \times 100\%$$

- Usually this is expressed as an absolute value (i.e., without any sign)

Approximate Error:

- used to measure the lack of precision of an estimate.
- Approximate (absolute) error
 $E_a = \text{Current approximation} - \text{Previous approximation}$
- Approximate relative error = $E_a / \text{Current app}$

- Approximate percent relative error = ϵ_a

$$= \frac{\text{Current approx.} - \text{Previous approx}}{\text{Current approx}} \times 100\%$$

- Usually this is also expressed as an absolute value (i.e., without any sign)

Exercise (HW 1)

The following sequence of estimates was obtained when a numerical method was applied to solve the equation:

$$x^4 - 5x - 7 = 0.$$

1.8254 1.9633 2.0121 2.0283 2.0335
2.0351 2.0356 2.0358

Calculate the four errors for these estimates, given that one of the roots of the equation is 2.0359.

Significant Figures

- Usually computation is repeated until the
 - approximate percent relative error ($|\epsilon_a|$) is within the pre-specified acceptable level (denoted by ϵ_s) i.e., until $|\epsilon_a| < \epsilon_s$
 - ϵ_s is usually determined by specifying the number of significant figures required.
 - If the number of significant figures required is at least **n**, then ϵ_s must be

$$(0.5 \times 10^{2-n})\%$$

Example:

- Suppose we wish to get the result of a computation correct to 3 significant figures i.e., $n = 3$
- Here we must fix ϵ_s as
$$\epsilon_s = 0.5 \times 10^{2-3} \%$$
$$= 0.05\%$$
- It means that we must improve our estimate until $|\epsilon_a|$ becomes less than 0.05

- Suppose our first two estimates are 1.23456 and 1.23785.

$$\begin{aligned} |\varepsilon_a| &= \left| \frac{1.23785 - 1.23456}{1.23785} \right| \times 100 \\ &= 0.265783 > 0.05 \end{aligned}$$

- Since $|\varepsilon_a|$ is more than ε_s , we must continue the iteration.

Suppose the third estimate is 1.23812.

Error is now estimated as

$$\begin{aligned} |\varepsilon_a| &= \left| \frac{1.23812 - 1.23785}{1.23812} \right| \times 100 \\ &= 0.021807 < 0.05 \end{aligned}$$

Now we see that $|\varepsilon_a|$ is less than ε_s .

Hence we have got the precision required and we can stop the iteration.

Problem

The Maclaurin series expansion for $\sin x$ is

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - + \dots$$

- (i) Starting with the simplest version $\sin x = x$, add terms one at a time to estimate $\sin(\pi/4)$.
- (ii) After each term is added, compute the true and approximate percent relative errors.
- (iii) From the calculator, determine the true value. Add terms until the absolute value of the approximate error falls below an error conforming to four significant figures.

Solution:

$$\pi/4 = 0.7853$$

True value of $\sin(\pi/4) = 0.707107$

Estimate	ε_a (%)	ε_t (%)
0.7853982	-	11.0720735
0.7046527	11.4589099	-----
0.7071430	-----	0.0051286
-----	0.0051727	0.0000441
0.7071068	0.0000443	0.0000002